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TECHNICAL SUMMARY

Wind–Solar Hybrid System

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ABSTRACT

Modeling a wind–solar hybrid with battery storage for reliable off-grid power in rural Nepal. Designed and modeled a hybrid renewable energy system that combines wind turbines, solar PV and battery storage, and proposed an optimal configuration for rural electrification.

Wind turbines	20 kW
Solar PV	15 kWp
Storage	Battery

1. Problem and motivation

Many rural communities in Nepal are far from the national grid. Wind and solar complement each other through the day and across seasons, but the system has to be sized and controlled carefully to cover the load reliably.

2. Methodology

- Designed and modeled the wind, solar and battery system
- Analyzed power sharing between wind, solar and storage under varying load and environmental conditions
- Evaluated performance, energy management and stability of the hybrid microgrid in MATLAB/Simulink

3. Figures

SAMPLE



Figure 1. Concept of the hybrid site: wind turbines, PV array, battery room and village load.

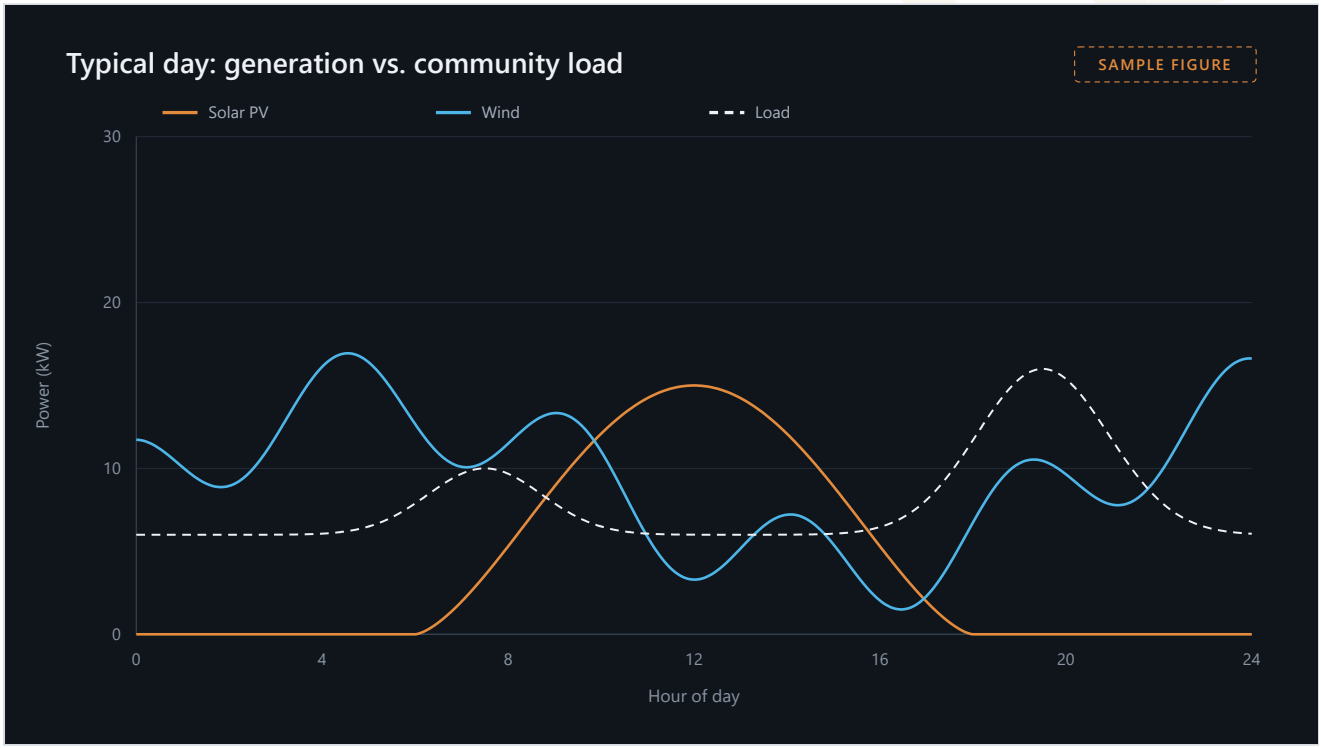


Figure 2. Typical day: solar and wind generation against community load.

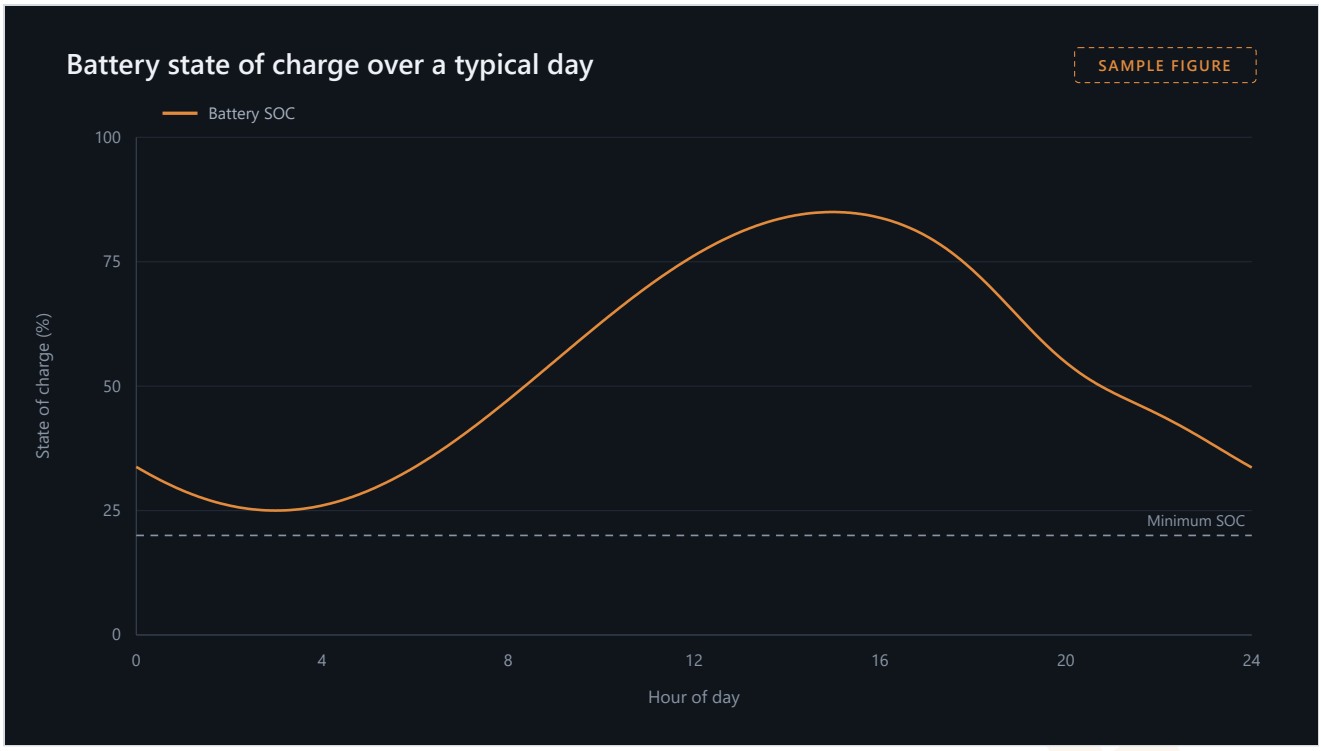


Figure 3. Battery state of charge over the same day.

4. Results

- Proposed an optimal configuration for rural electrification, improving reliability and use of renewable resources
- Published in the Journal of the Institute of Engineering, vol. 15, no. 3, pp. 360–367 (2019)

5. Team and contribution

Contribution: Co-author. Contributed system modeling, power-sharing analysis and performance evaluation in MATLAB/Simulink.

Tools: MATLAB/Simulink, Load analysis, Energy management

6. References

1. Paper (DOI): https://doi.org/10.3126/jie.v15i3.32223

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